

Jayesh Prasad Anandan

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PROFESSIONAL SUMMARY

Senior Cloud Data Engineer with strong expertise in Azure Data Factory, Synapse Analytics, Databricks, and enterprise-scale pipeline orchestration. Skilled in building robust ingestion frameworks, optimizing data warehousing operations, and resolving critical production failures across statewide public-health systems. Demonstrated success in cross-team collaboration, CI/CD automation, access/security governance, and enhancing platform reliability. Backed by a Master's degree in Data Science and cloud certifications with a focus on delivering high-quality, scalable data solutions that support decision-making at scale.

CORE COMPETENCIES

Cloud Platforms: Azure (Fabric, Databricks, Data Factory, Synapse, Data Lake, DevOps, Logic Apps), AWS (S3, Glue, Redshift, Bedrock, SageMaker)

Data Engineering: ETL/ELT, Data Warehousing, Data Modeling, Batch & Real-time Processing, Data Governance

Programming & Databases: Python, PySpark, Scala, SQL, R, Java, C++, Unix Shell, MySQL, PostgreSQL, MongoDB

Frameworks & Tools: Apache Airflow, Databricks, Hadoop, Spark, Kafka, Tensorflow, PyTorch, MuleSoft

Analytics & Visualization: Tableau, Dataiku, Alteryx, Google Analytics, Looker Studio, Excel, ArcGIS

PROFESSIONAL EXPERIENCE

Senior Data Engineer - Contract

HSK Technologies Inc - Indiana Department of Health

November 2024 – Present

Indianapolis, IN

- Restored a statewide-critical Record Linkage system by diagnosing missing JAR dependencies in the Databricks pre-execution chain, tracing failures across init scripts and environment configs, and fully re-enabling linkage workflows for Trauma Registry, Children and Hoosier Immunization Registry Program (CHIRP) and Vital Records
- Engineered and enhanced 20+ mission-critical ingestion pipelines across CHIRP, Mom's Help Line (MHL) previously known as My Healthy Baby (MHB), Hospital Discharge, Indiana Health Information Exchange (IHIE), Tick, Flu, Monkeypox, and Census Select & Arrange (CENSA), enabling timely statewide public-health reporting and analytics
- Improved reliability and performance of IHIE's historical file pipeline by implementing TAR GZIP compression, reducing ingestion runtime from 10+ hours to 1h 45m, an 83% reduction in runtime and shrinking file size from 22.9GB to 3.7GB
- Led large-scale ingestion and transformation for Hospital Discharge datasets (2015–2024), resolving Databricks logic errors, rebuilding ingestion flows, and enabling consistent quarterly and annual reporting
- Designed end-to-end reporting solutions by integrating ADF/Synapse pipelines with Tableau, Power BI, and ArcGIS dashboards to deliver accurate, timely, and visually intuitive analytics across multiple public health domains
- Designed and maintained large-scale data ingestion for CHIRP Immunization datasets, creating Synapse schemas and expanding ADF ingestion to incorporate 15+ new tables requested by Immunization program leadership
- Built resilient ADF pipelines ingesting on-prem Oracle data, CHIRP Replica tables, VisionLink program data, K-Drive quarterly MHB files, and manual program submissions, ensuring end-to-end operational continuity

- Modernized and stabilized MHB ingestion by updating VisionLink schemas, syncing program data model changes, adding new Client and Interaction fields, and fixing multiple pipeline failures triggered by source system updates
- Improved DW Synapse reliability by implementing table statistics updates, fixing SPROC logic, correcting schema mismatches, and resolving deployment blockers such as connectivity failures and package issues
- Improved cross-environment consistency by updating ADF linked services and integration runtimes across Dev and Prod, ensuring accurate compute routing after infrastructure-level changes
- Resolved multiple production failures across COVID Dashboard, Tick, Flu, and Wastewater pipelines by repairing SPROCs, adjusting ingestion logic, or updating Databricks dependencies
- Fixed CHIRP ingestion failures caused by Synapse memory pool exhaustion by redesigning copy activity sequencing for heavy tables—restoring stable ingestion of large datasets
- Repaired MHB VisionLink ingestion failures related to Oracle idle timeout errors, column renames, and schema changes, ensuring pipeline consistency and uninterrupted delivery to Program leadership
- Strengthened CI/CD release workflows by contributing to ADF Publish Automation and resolving deployment warnings involving Ubuntu image changes, Az module versions, and key-vault encryption policies
- Troubleshoot and resolved DevOps pipeline failures (sqlpackage path errors, environment permission issues, container registry access blocks), ensuring smooth deployment for Synapse, Databricks, and ADF assets
- Enabled secure and compliant operations by provisioning Synapse, ADF, and Storage access for IZ, Trauma, Chronic Disease, BI, & ACE teams; resolved VDI issues, shared-drive problems, and permission mismatches
- Supported Program areas by resolving schema-update needs, storage limitations, copy activity failures, linked service issues, and DataOps cadence adjustments
- Updated critical technical documentation across MHB, Table Statistics, Vital Records schemas, GoAnywhere processes, DevOps standards, and ACE onboarding SOPs—ensuring operational clarity and audit readiness

Part-Time Marketing Data Analyst

UCM, Indiana University

March 2025 – June 2025 & April 2023 – May 2024

Bloomington, IN

- Implemented comprehensive analytics tracking across 10 IU campuses using Google Analytics, Google Tag Manager, and Search Console, enabling targeted marketing strategies
- Reduced reporting time from 36+ hours to 2 hours by building automated data extraction system for domain migration of 750 campus websites using Selenium and Python
- Managed migration of 500+ tags and triggers within Google Tag Manager, ensuring seamless domain transition with zero downtime

Senior Consultant

Hoosier Community Network

July 2024 – December 2024

Bloomington, IN

- Led 25+ data analysts and developers across 8+ client projects, ensuring on-time delivery and high satisfaction through effective resource allocation and performance management
- Engineered web scraping solution using Python and BeautifulSoup to extract 20,000+ web pages from open-source platforms, storing cleaned data in PostgreSQL for analytics
- Automated ETL workflows using Alteryx Designer and Tableau Prep, reducing manual data handling time by 28% across multiple data sources

Data Scientist

Project 990 Inc.

August 2024 – October 2024

Bloomington, IN

- Improved data quality by 47% by developing custom Python package integrated with Alteryx workflows to validate and cleanse IRS 990 nonprofit financial data
- Optimized data ingestion pipeline reducing processing time by 32%, enabling terabyte-scale data ingestion into PostgreSQL for business-critical nonprofit analytics

Data Engineer

TATA Consultancy Services Limited

November 2020 – July 2022

Chennai, India

- Designed and implemented metadata driven AWS-based ETL pipelines with PySpark and Apache Airflow, orchestrating workflows for financial data processing and dashboard creation
- Reduced data ingestion time by 23% by developing a metadata-driven framework for financial data pipelines using Amazon EMR and EC2, eliminating manual configuration
- Architected Flask-based metadata generation API that automatically extracted file schemas and properties, eliminating manual cataloging bottlenecks and improving ingestion pipeline throughput by 50%
- Optimized data transfer efficiency by 68% by accelerating JDBC-based data retrieval and write operations with connection pooling and batch processing in Python
- Led team of 5 junior data engineers, mentoring on AWS best practices, PySpark optimization, and data pipeline design

Machine Learning Engineer Intern

TATA Consultancy Services Limited

December 2019 – April 2020

Chennai, India

- Recognized as Runner-Up in Framework Development Category by Global Technology Division for pioneering text-to-video content generation framework using native coding tools, pre-dating modern AI solutions
- Increased customer satisfaction by 21% by developing NLP-based text generation model using TensorFlow and Keras to automate user story creation
- Pioneered a Mobile Application for Object detection of Designs and Patterns in custom made products using Keras and Tensorflow
- Formulated a surveillance application to detect exam malpractice from CCTV footage using PoseNet

KEY PROJECTS

Psychological Feature Space using Vision Transformers — Python, Keras, Computer Vision 2024

- Achieved a 50% average correlation coefficient across 8 MDS dimensions, by predicting Multi-Dimensional Scaling (MDS) values through image classification to understand human categorization
- Improved model accuracy and performance, by training and fine-tuning an ensemble of 10 deep learning models, including DenseNet and Vision Transformers

EDUCATION

Master of Science in Data Science

Indiana University

May 2024

Bloomington, IN

Bachelor of Engineering in Electrical & Electronics Engineering

Anna University

May 2020

Chennai, Tamil Nadu, India

CERTIFICATIONS

AWS Certified Data Engineer – Associate — AWS Cloud

August 2024 - August 2027

Microsoft Certified: Azure Data Engineer Associate — Microsoft Azure

March 2025 - March 2026

Anypoint Platform Development: Fundamentals — Salesforce

February 2025

Alteryx Designer Core Certification — Alteryx

July 2024

Deep Learning Specialization Certificate — DeepLearning.AI

November 2020